Name → Ankush Saha Regd. → 2341019012 Serial → 51 Section → L2

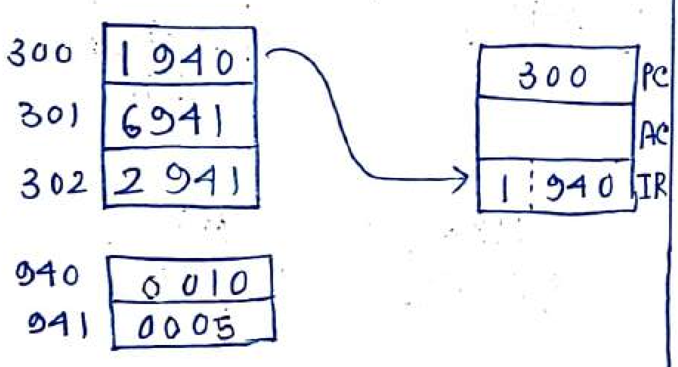
DPOS Assignment - 1

1) 0001 → Load AC from memory
0010 → Store AC to memory
0110 → Subtract the value of AC from memory

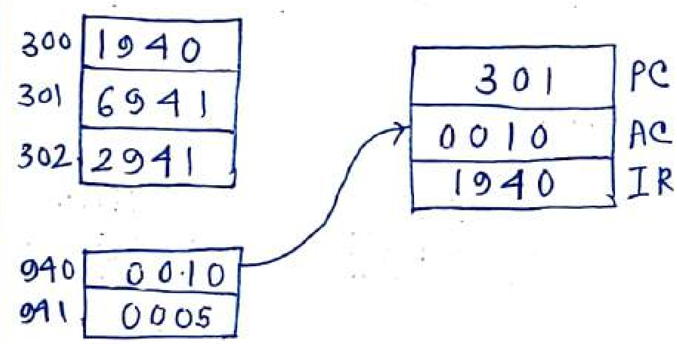
Show the Program execution, assume 940 & 941 has value 10 & 5

Step 1

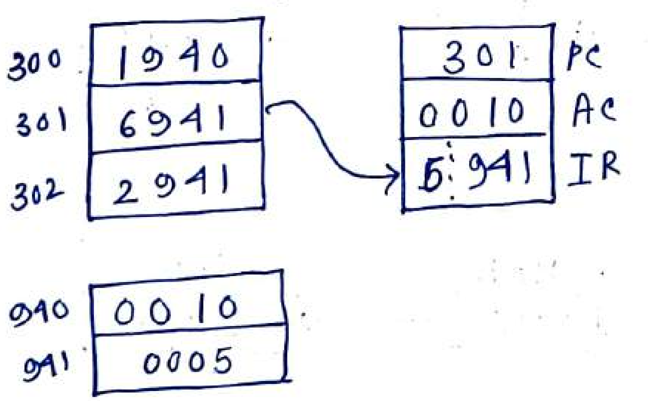
Fetch stage

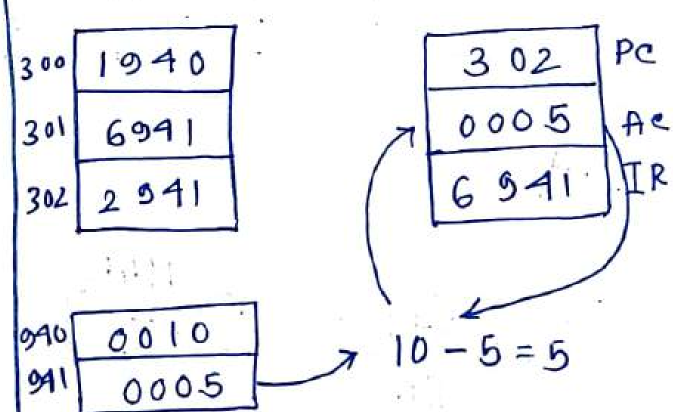


Execute stage

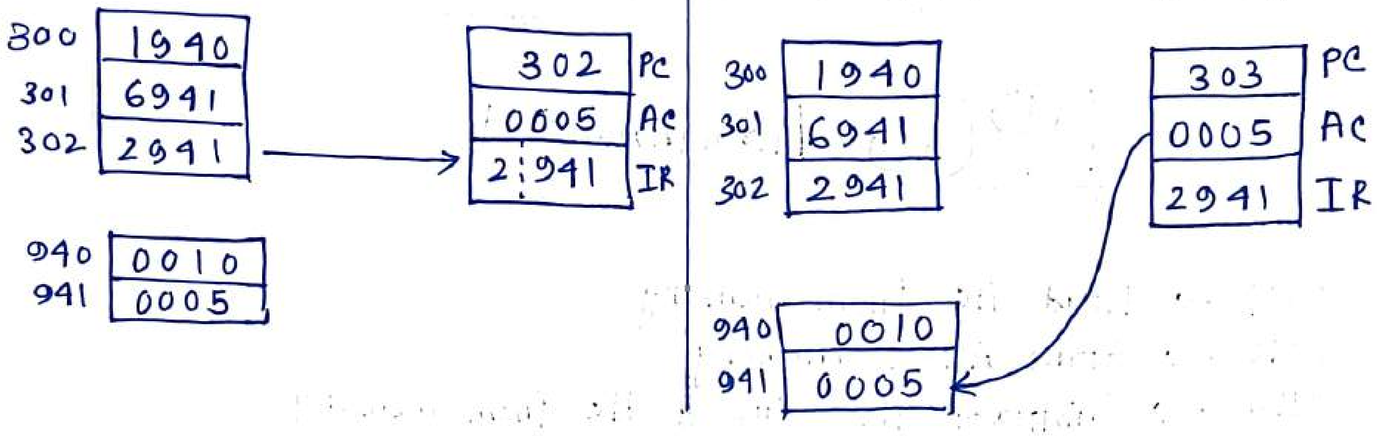


Step 2



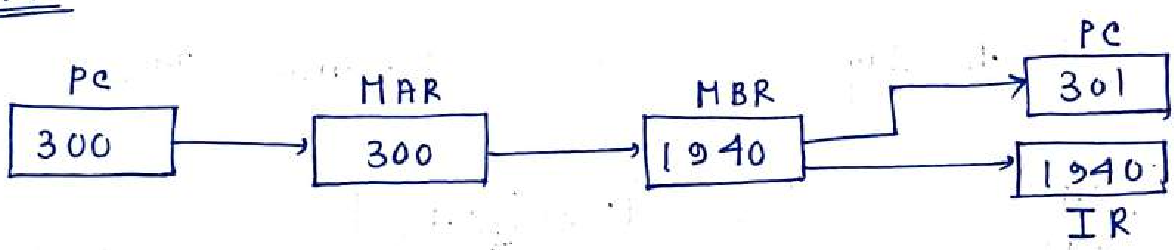


Step 3

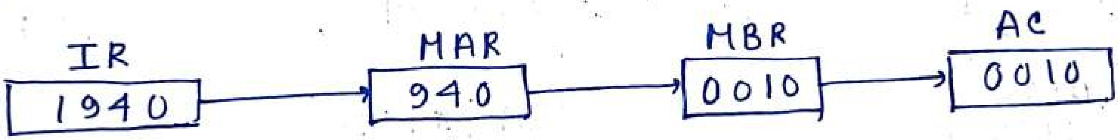


2) Expand the above question to show the use of MAR & MBR at each step

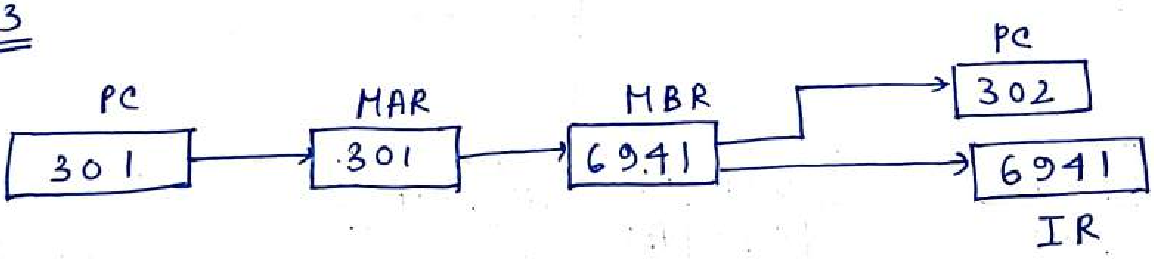
Step 1



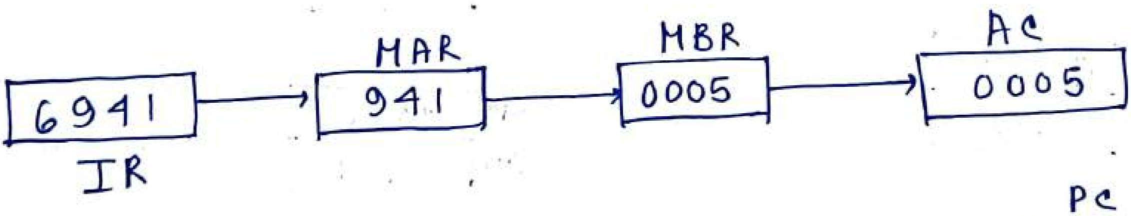
Step 2



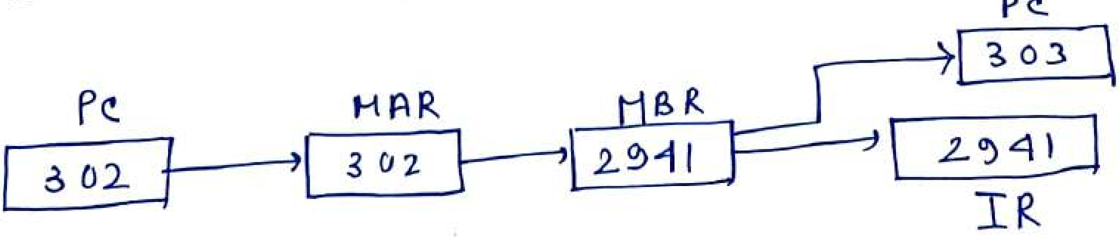
Step 3



Step 4



Step 5



Step 6

IR		MAR		AC		MBR
2941	→	941	→	0005	→	0005

3) Cache access time is 50 ns, Main memory access time = 550 ns, Read involves 80% of all operations & its hit ratio is 0.9, Avg. access time of the system

Avg. access time for read = $(0.8 \times 50) + (0.2)(550)$

$= 100$ ns

Avg. access time = $T_{avg\,read} \times P_{read} + T_{write} \times P_{write}$

$= (100) \times (0.8) + (0.2)(550)$

$= 190$ ns

Ans

4) Find CPU utilization, memory utilization & Throughput in case of Uniprograming & Multiprograming system.

Uniprogramming

$$\text{CPU time} = \frac{70\% \text{ of } 10 + 10\% \text{ of } 20 + 10\% \text{ of } 15}{45}$$

$$= \frac{7 + 2 + 1.5}{45}$$

$$= 0.2333 = 23.33\%$$

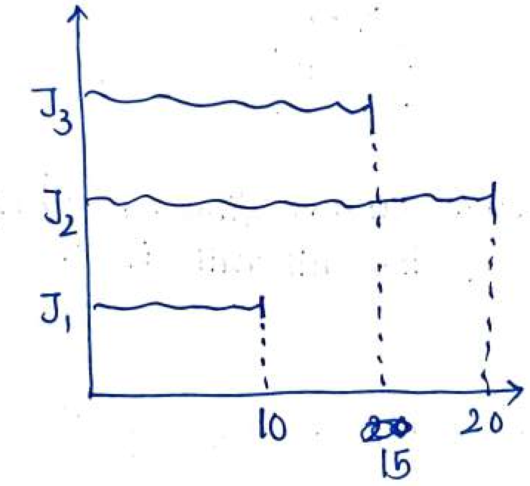
$$\text{Throughput} = \frac{3}{45} \times 60 = 4 \text{ Jobs/hr}$$

$$\text{Memory utilization} = \frac{150 \times 10 + 100 \times 20 + 125 \times 15}{400 \times 45}$$

$$= 0.2986$$

$$= 29.86\%$$

Multiprogramming



$$\text{CPU time/utilisation} = \frac{(0.7 \times 10) + (0.1 \times 20) + (0.1 \times 15)}{20}$$

$$= \frac{7 + 2 + 1.5}{20}$$

$$= 0.525$$

$$= 52.5\%$$

$$\text{Throughput} = 3/20 \times 60 = 9 \text{ Jobs/hr}$$

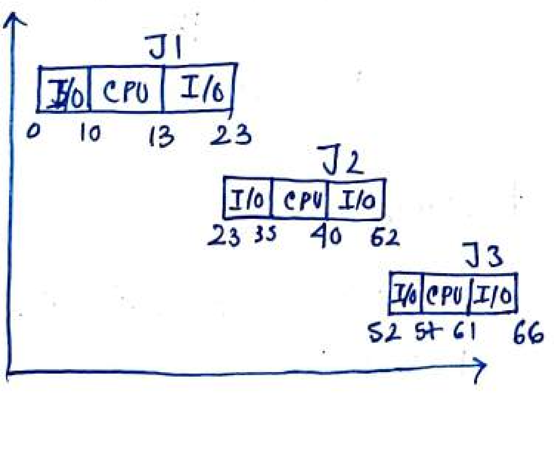
$$\text{Memory utilisation} = \frac{(150 + 100 + 125) \times 10 + (100 + 125) \times \cancel{8}5 + (100 \times 5)}{400 \times 20}$$

$$= \frac{3750 + 1125 + 500}{8000}$$

$$= 0.6718 = 67.18\%$$

5) Find CPU utilisation for uniprogramming & multiprogramming in batch OS

Uniprogramming

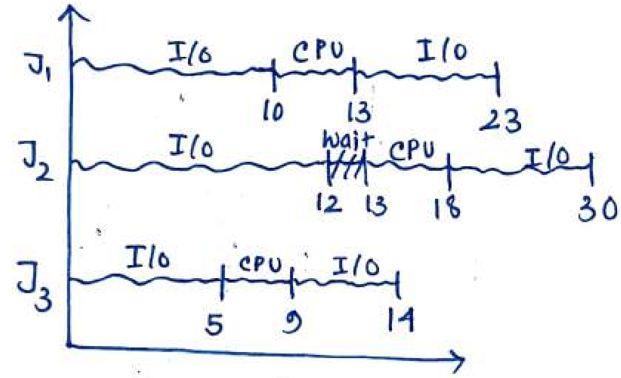


$$\therefore \text{CPU utilisation} = \frac{3 + 5 + 4}{66}$$

$$= 0.181$$

$$= 18.1\%$$

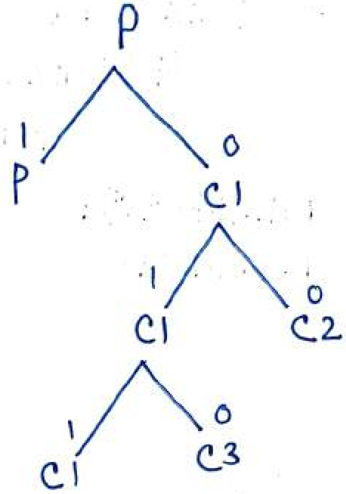
5 Multiprogramming



$$\text{CPU utilization} = \frac{3+4+5}{30} = 0.4 = 40\%$$

6) (a)

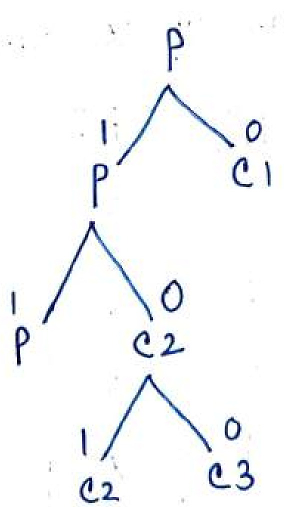
```
int main(){
    Pid_t, c1, c2;
    c2 = 0;
    c1= fork();
    if (c1 == 0){
        c2 = fork();
    }
    if (c2 >0){
        fork();
    }
    Printf ("1");
    return 0;
}
```



Total Processes = 4
Total child Processes = 3

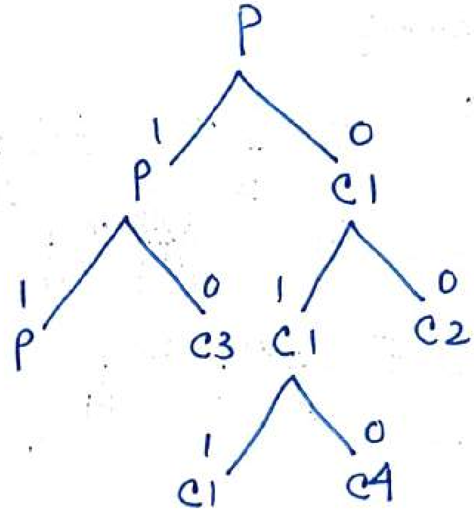
(b)

```
int main(){
    Pid_t c1, c2 = 1;
    c1 = fork();
    if (c1 != 0)
        c2 = fork();
    if (c2 == 0)
        fork();
    Printf ("1");
    return 0;
}
```



Total Processes = 4

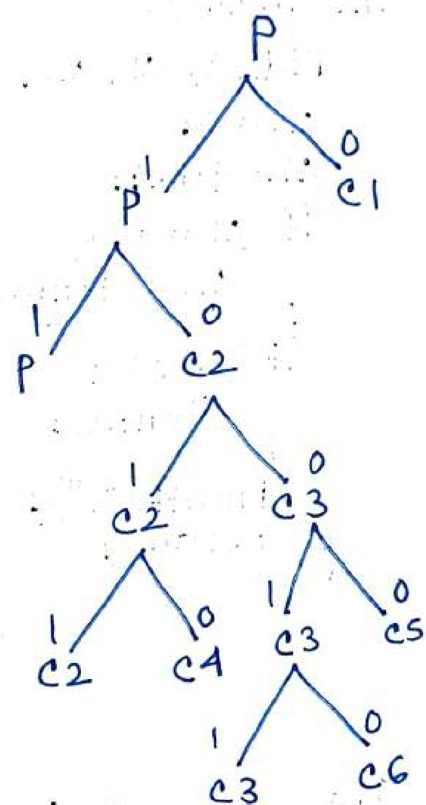
(c)
```
int main(){
    if (fork() || fork())
        fork();
    printf("1");
    return 0;
}
```



∴ Total Processes = 5

(d)
```
int main(){
    if (fork() && (!fork()))
        if (fork() || fork())
            fork();

    printf("2");
    return 0;
}
```



∴ Total Processes = 7

7) Find value of x s.t. avg. WT = 1 ms

Process	AT	BT
P_1	0	3
P_2	1	1
P_3	3	3
P_4	4	x

RQ: $\cancel{P_1}^{\cancel{3}\cancel{7}\cancel{1}}$ $\cancel{P_2}$ $\cancel{P_3}^{\cancel{3}}$ $\cancel{P_4}^{\cancel{x}}$

case 1 $(x < 3)$

gantt chart :

P_1	P_2	P_1	P_1	P_4	P_3

0, 1, 2, 3, 4, $4+x$, $7+x$

CT	TAT	WT
4	4	1
2	1	0
$7+x$	$4+x$	$1+x$
$4+x$	x	0

$\therefore$ avg. WT $= \dfrac{1+1+x}{4}$

$\Rightarrow 1 = \dfrac{2+x}{4}$

$\Rightarrow 4 = 2+x$

$\Rightarrow x = 2$ ms

case 2 $(x \geq 3)$

gantt chart :

P_1	P_2	P_1	P_1	P_3	P_4

0, 1, 2, 3, 4, 7, $7+x$

CT	TAT	WT
4	4	1
2	1	0
7	4	1
$7+x$	$3+x$	3

avg. WT $= \dfrac{1+1+3}{4}$

$\Rightarrow 1 = 5/4$

$\Rightarrow 5 \neq 4$

$\therefore$ False case

$\therefore$ $\boxed{x = 2\text{ ms}}$ Ans

8)

FCFS

Process	AT	BT	CT	TAT	WT
P_1	0	11	11	11	0
P_2	0	8	19	19	11
P_3	12	2	43	31	29
P_4	2	6	25	23	17
P_5	9	16	41	32	16

RQ: P_1 P_2 P_4 P_5 P_3

gantt chart:

P_1	P_2	P_4	P_5	P_3

0 11 19 25 41 43

avg. TAT = $116/5 = 23.2$

avg. WT = $73/5 = 14.6$

SJF

Process	AT	BT	CT	TAT	WT
P_1	0	11	27	27	16
P_2	0	8	8	8	0
P_3	12	2	16	4	2
P_4	2	6	14	12	6
P_5	9	16	43	34	18

RQ: $\cancel{P_1}$ $\cancel{P_2}$ $\cancel{P_4}$ $\cancel{P_5}$ $\cancel{P_3}$

gantt chart:

P_2	P_4	P_3	P_1	P_5

0 8 14 16 27 43

avg. TAT = $85/5 = 17$

avg. WT = $42/5 = 8.4$

Ans

SRTF

Process	AT	BT	CT	TAT	WT
P_1	0	11	27	27	16
P_2	0	8	8	8	0
P_3	12	2	16	4	2
P_4	2	6	14	12	6
P_5	9	16	43	34	18

RQ: $\cancel{P_1}^{11}$ $P_2^{8\cancel{6}}$ $\cancel{P_4}^{\cancel{6}\cancel{3}\cancel{2}}$ $\cancel{P_5}^{16}$ $\cancel{P_3}^{2}$

gantt chart:

P_2	P_2	P_4	P_4	P_4	P_3	P_1	P_5

0 2 8 9 12 14 16 27 43

avg. TAT = $85/5 = 17$

avg. WT = $42/5 = 8.4$

Priority based non Preemption

Process	AT	BT	Priority	CT	TAT	WT
P_1	0	11	1	11	11	0
P_2	0	8	0	43	43	35
P_3	12	2	3	29	17	15
P_4	2	6	2	35	33	27
P_5	9	16	4 (high)	27	18	2

RQ: $\cancel{P_1}$ P_2 P_4 $\cancel{P_5}$ $\cancel{P_3}$

gantt chart:

P_1	P_5	P_3	P_4	P_2

0 11 27 29 35 43

avg. TAT = $122/5 = 24.4$

avg. WT = $79/5 = 15.8$

Preemptive Priority

Process	AT	BT	Priority	CT	TAT	WT
P_1	0	11	1	35	35	24
P_2	0	8	0	43	43	35
P_3	12	2	3	27	15	13
P_4	2	6	2	8	6	0
P_5	9	16	4	25	16	0

RQ: $\cancel{P_1}^{11\ 9\ 8}$ $\cancel{P_2}^{8}$ $\cancel{P_4}^{6}$ $\cancel{P_5}^{16}$ $\cancel{P_3}^{2}$

gantt chart:

P_1	P_4	P_1	P_5	P_3	P_1	P_2

0 2 8 9 25 27 35 43

avg. TAT = 115/5 = 23

avg. WT = 72/5 = 14.4

Highest Response ratio next

Process	AT	BT	CT	TAT	WT
P_1	0	11	11	11	0
P_2	0	8	27	27	19
P_3	12	2	19	7	5
P_4	2	6	17	15	9
P_5	9	16	43	34	18

RQ: $\cancel{P_1}$ $\cancel{P_2}$ $\cancel{P_4}$ $\cancel{P_5}$ $\cancel{P_3}$

avg. TAT = 94/5 = 18.8

avg. WT = 51/5 = 10.2

gantt chart:

P_1	P_4	P_3	P_2	P_5

0 11 17 19 27 43

P_1 & P_2

$P_1 \longrightarrow W = 0-0 \Rightarrow \frac{0+11}{11} = 1$ (FCFS)

$P_2 \longrightarrow W = 0-0 \Rightarrow \frac{0+8}{8} = 1$

P_2, P_4, P_5

$P_2 \longrightarrow W = 11-0 \Rightarrow \frac{11+8}{8} = 2.375$

$P_4 \longrightarrow W = 11-2 \Rightarrow \frac{9+6}{6} = 2.5$ (highest)

$P_5 \longrightarrow W = 11-9 \Rightarrow \frac{2+16}{16} = 1.125$

P_2, P_5, P_3

$P_2 \longrightarrow W = 17-0 \Rightarrow \frac{17+8}{8} = 3.125$

$P_5 \longrightarrow W = 17-9 \Rightarrow \frac{8+16}{16} = 1.5$

$P_3 \longrightarrow W = 17-12 \Rightarrow \frac{5+2}{2} = 3.5$ (highest)

P_2, P_5

$P_2 \longrightarrow W = 19-0 \Rightarrow \frac{19+8}{8} = 3.375$ (highest)

$P_5 \longrightarrow W = 19-9 \Rightarrow \frac{10+16}{16} = 1.625$

RR (TQ=2)

Process	AT	BT	CT	TAT	WT
P_1	0	11	33	33	22
P_2	0	8	28	28	20
P_3	12	2	20	8	6
P_4	2	6	22	20	14
P_5	9	16	43	34	18

RQ: $\cancel{P_1}$ $\cancel{P_2}$ $\cancel{P_4}$ $\cancel{P_1}$ $\cancel{P_2}$ $\cancel{P_4}$ $\cancel{P_1}$ $\cancel{P_5}$ $\cancel{P_2}$ $\cancel{P_3}$ $\cancel{P_4}$ $\cancel{P_1}$ $\cancel{P_5}$ $\cancel{P_2}$ $\cancel{P_1}$ $\cancel{P_5}$ $\cancel{P_1}$ $\cancel{P_5}$

gantt chart:

P_1	P_2	P_4	P_1	P_2	P_4	P_1	P_5	P_2	P_3	P_4	P_1	P_5	P_2	P_1	P_5	P_1
0–2	2–4	4–6	6–8	8–10	10–12	12–14	14–16	16–18	18–20	20–22	22–24	24–26	26–28	28–30	30–32	32–34

P_5
33 – 43

avg. TAT = $123/5$ = 24.6

avg. WT = $80/5$ = 16

$\therefore$ SJF/SRTF produces the min. waiting time

9)

FCFS

Process	AT	BT	CT	TAT	WT
P_1	0	4	4	4	0
P_2	0	2	6	6	4
P_3	1	3	9	8	5
P_4	2	2	11	9	7

RQ: $\cancel{P_1}$ $\cancel{P_2}$ $\cancel{P_3}$ $\cancel{P_4}$

gantt chart:

P_1	P_2	P_3	P_4
0 – 4	4 – 6	6 – 9	9 – 11

avg. TAT = $27/4$ = 6.75

avg. WT = $16/4$ = 4

SJF

Process	AT	BT	CT	TAT	WT
P_1	0	4	11	11	7
P_2	0	2	2	2	0
P_3	1	3	7	6	3
P_4	2	2	4	2	0

RQ: $\cancel{P_1}$ $\cancel{P_2}$ $\cancel{P_3}$ $\cancel{P_4}$

avg. TAT = $21/4$ = 5.25

avg. WT = $10/4$ = 2.5

gantt chart:

P_2	P_4	P_3	P_1
0 – 2	2 – 4	4 – 7	7 – 11

~~P5~~
~~33~~ ~~43~~

~~avg. TAT = 123/5 = 24.6~~

~~avg. WT = 80/5 = 16~~

SRTF

Process	AT	BT	CT	TAT	WT
P_1	0	4	11	11	7
P_2	0	2	2	2	0
P_3	1	3	7	6	3
P_4	2	2	4	2	0

RQ : ~~P_1^{4}~~ ~~$P_2^{2\,1}$~~ ~~P_3^{3}~~ ~~P_4^{2}~~

avg. TAT = 21/4 = 5.25

avg. WT = 10/4 = 2.5

gantt chart:

P_2	P_2	P_4	P_3	P_1
0 – 1	1 – 2	2 – 4	4 – 7	7 – 11

Priority non Preemptive

Process	AT	BT	Priority	CT	TAT	WT
P_1	0	4	3	9	9	5
P_2	0	2	1 (high)	2	2	0
P_3	1	3	2	5	4	1
P_4	2	2	4	11	9	7

RQ : ~~P_1~~ ~~P_2~~ ~~P_3~~ ~~P_4~~

gantt chart:

P_2	P_3	P_1	P_4
0 – 2	2 – 5	5 – 9	9 – 11

avg. TAT = 24/4 = 6

avg. WT = 13/4 = 3.25

Priority with preemption

Process	AT	BT	Priority	CT	TAT	WT
P_1	0	4	3	9	9	5
P_2	0	2	1 (high)	2	2	0
P_3	1	3	2	5	4	1
P_4	2	2	4	11	9	7

RQ: $\cancel{P_1}$ $\cancel{P_2}$ $\cancel{P_3}$ $\cancel{P_4}$

gantt chart:

P_2	P_2	P_3	P_1	P_4
0–1	1–2	2–5	5–9	9–11

avg. TAT = $24/4 = 6$

avg. WT = $13/4 = 3.25$

Highest response ratio next

Process	AT	BT	CT	TAT	WT
P_1	0	4	4	4	0
P_2	0	2	6	6	4
P_3	1	3	11	10	7
P_4	2	2	8	6	4

RQ: $\cancel{P_1}$ $\cancel{P_2}$ P_3 P_4

avg. TAT = $\frac{26}{4} = 6.5$

avg. WT = $15/4 = 3.75$

gantt chart:

P_1	P_2	P_4	P_3
0–4	4–6	6–8	8–11

P_1, P_2

$P_1 \rightarrow W = 0-0 \Rightarrow \frac{0+4}{4} = 1$

$P_2 \rightarrow W = 0-0 \Rightarrow \frac{0+2}{2} = 1$

P_2, P_3, P_4

$P_2 \rightarrow W = 4-0 \Rightarrow \frac{4+2}{2} = 3$ (high)

$P_3 \rightarrow W = 4-1 \Rightarrow \frac{3+3}{3} = 2$

$P_4 \rightarrow W = 4-2 \Rightarrow \frac{2+2}{2} = 2$

P_3, P_4

$P_3 \rightarrow W = 6-1 \Rightarrow \frac{5+3}{3} = 2.67$

$P_4 \rightarrow W = 6-2 \Rightarrow \frac{4+2}{2} = 3$ (high)

RR (TQ=2)

Process	AT	BT	CT	TAT	WT
P_1	0	4	10	10	6
P_2	0	2	4	4	2
P_3	1	3	11	10	7
P_4	2	2	8	6	4

avg. TAT = 30/4
= 7.5

avg. WT = 19/4
= 4.75

RQ: $\cancel{P_1^4}$ $\cancel{P_2^2}$ $\cancel{P_3^3}$ $\cancel{P_4^2}$ $\cancel{P_1^2}$ $\cancel{P_3^1}$

gantt chart:

P_1	P_2	P_3	P_4	P_1	P_3	
0	2	4	6	8	10	11

∴ SJF / SRTF produces the minimum waiting time

10) Find avg. WT & avg. TAT

Process	AT	BT	CT	TAT	WT
P_1	0	8	17	17	9
P_2	0	22	42	42	20
P_3	0	4	23	23	19
P_4	0	12	46	46	34

$Q_1 \longrightarrow$ RR (TQ = 3) {high}

$Q_2 \longrightarrow$ RR (TQ = 5)

$Q_3 \longrightarrow$ FCFS

avg. TAT = 128/4 = 32

avg. WT = 82/4 = 20.5

RQ_1: $\cancel{P_1^8}$ $\cancel{P_2^{22}}$ $\cancel{P_3^4}$ $\cancel{P_4^{12}}$

RQ_2: $\cancel{P_1^5}$ $\cancel{P_2^{19}}$ $\cancel{P_3^1}$ $\cancel{P_4^9}$

RQ_3: $\cancel{P_2^{14}}$ $\cancel{P_4^4}$

gantt chart:

P_1	P_2	P_3	P_4	P_1	P_2	P_3	P_4	P_2	P_4	
0	3	6	9	12	17	22	23	28	42	46